Sales & Profitability: End-to-End Analysis

Capstone Project - Data Visualization & Analysis (Python, SQL, Pandas, Excel, Power BI/Tableau)

1. Executive summary

The business earned **$118.7M** in net sales and **$16.9M** in profit (**14.2%** margin) across 700 transactions between Sep 2013 and Dec 2014. Profit is concentrated: **Government** delivers 44% of sales and 67% of profit. The biggest leak is the **Enterprise** segment, which sells $19.6M but loses $0.61M, and **discounting**, which cuts margin from 21.9% (no discount) to 9.1% (High band).

2. Data & cleaning (Python)

- Dataset: 700 rows x 16 columns - segment, country, product, discount band, units, prices, gross sales, discounts, sales, COGS, profit, date.
- Column names standardised (the *Sales* column had a leading space); country 'United States of America' shortened to 'USA'.
- **53 blank Discount Band values** - all have Discounts = 0, so they mean 'no discount' and were labelled **None** (not guessed).
- No duplicate rows. Integrity checks passed: Gross - Discounts = Sales, Sales - COGS = Profit, Units x Price = Gross Sales, Date = Month/Year.
- Added features: profit margin %, discount %, month (YYYY-MM), loss flag. Large-deal profit outliers were kept - they are real deals, not errors.
- **Caveat:** 2013 only has Sep-Dec while 2014 has 12 months, so raw year totals are not comparable. Like-for-like (Sep-Dec) is used instead.

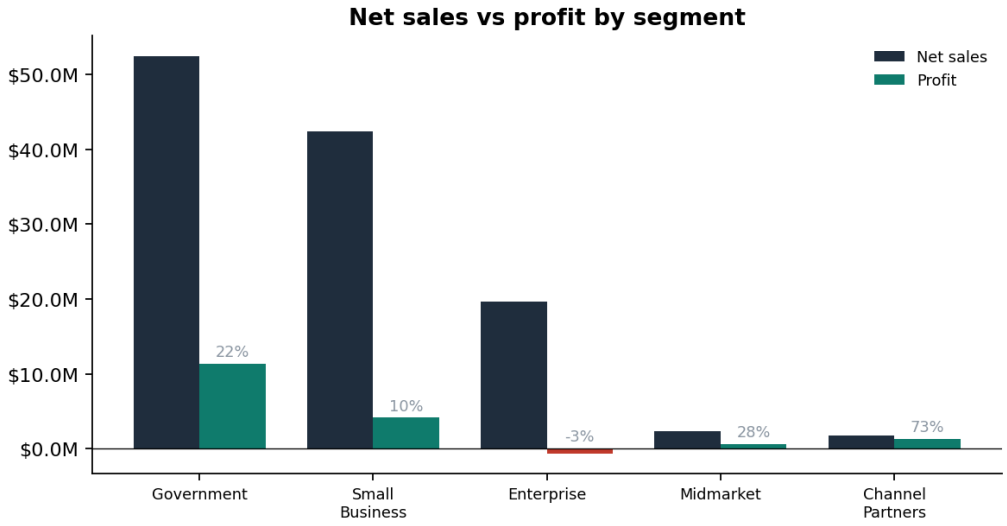
3. Method

Each question was answered twice - once in **SQL** (12 queries in *sql/queries.sql*, run on SQLite) and once in **pandas** - and the results were cross-checked in code. Summaries were then rebuilt in **Excel** with live SUMIFS formulas and charts, and visualised in a **BI dashboard**.

4. Key findings

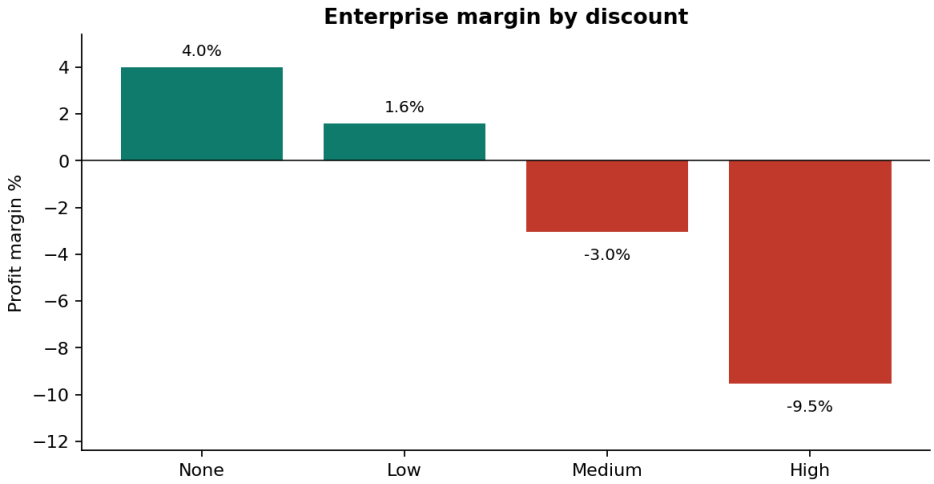
4.1 Segment performance

Segment	Net sales	Profit	Margin %	Share of profit %
Government	$52.50M	$11.39M	21.7	67.4
Small Business	$42.43M	$4.14M	9.8	24.5
Channel Partners	$1.80M	$1.32M	73.1	7.8
Midmarket	$2.38M	$0.66M	27.7	3.9
Enterprise	$19.61M	-$0.61M	-3.1	-3.6



- Government: 44% of sales, 67% of profit, 21.7% margin - the core of the business.
- Channel Partners: 73% margin but only $1.8M sales - high quality, low volume.
- Small Business: $42.4M sales but only 9.8% margin - volume without much profit.
- **Enterprise: negative margin (-3.1%); all 58 loss-making deals are Enterprise.**

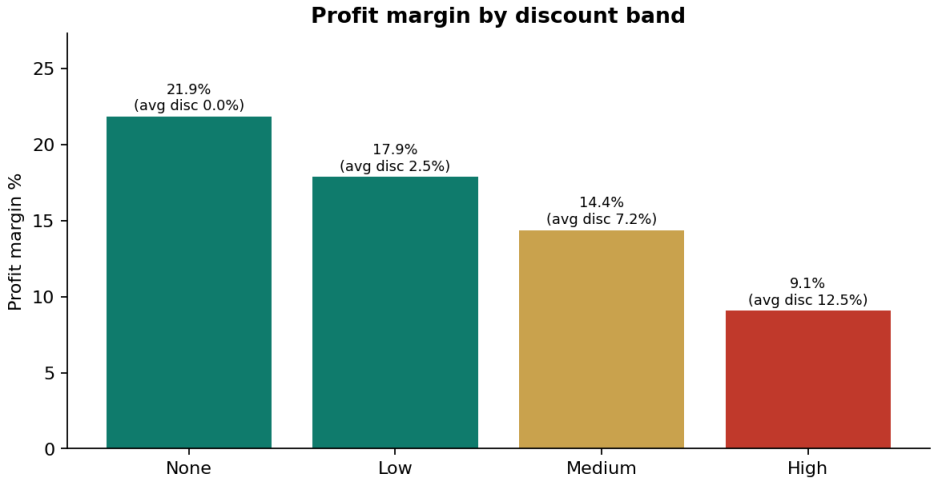
4.2 Why Enterprise loses money



- Enterprise sale price is 125 per unit against a unit cost of 120 - a gross margin of only **4%** before any discount.
- Discounts range 5-15% on the loss-making deals, so even a Low/Medium discount can wipe out the margin. High-band Enterprise deals run at -9.5%.
- Total losses on these 58 deals: **$0.78M**. Fixing Enterprise pricing is the single clearest profit lever.

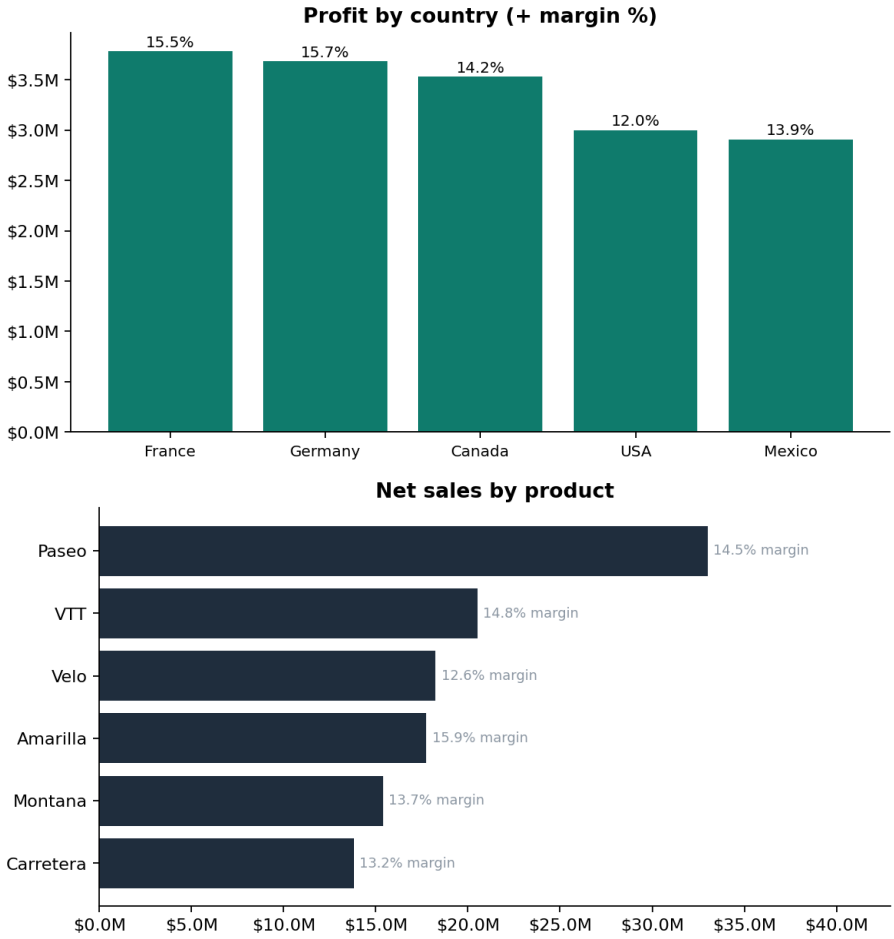
4.3 Discounts erode margin

Discount band	Deals	Avg discount %	Net sales	Margin %	Loss deals
None	53	0.0	$7.94M	21.9	0
Low	160	2.5	$34.63M	17.9	0
Medium	242	7.2	$38.78M	14.4	25
High	245	12.5	$37.37M	9.1	33



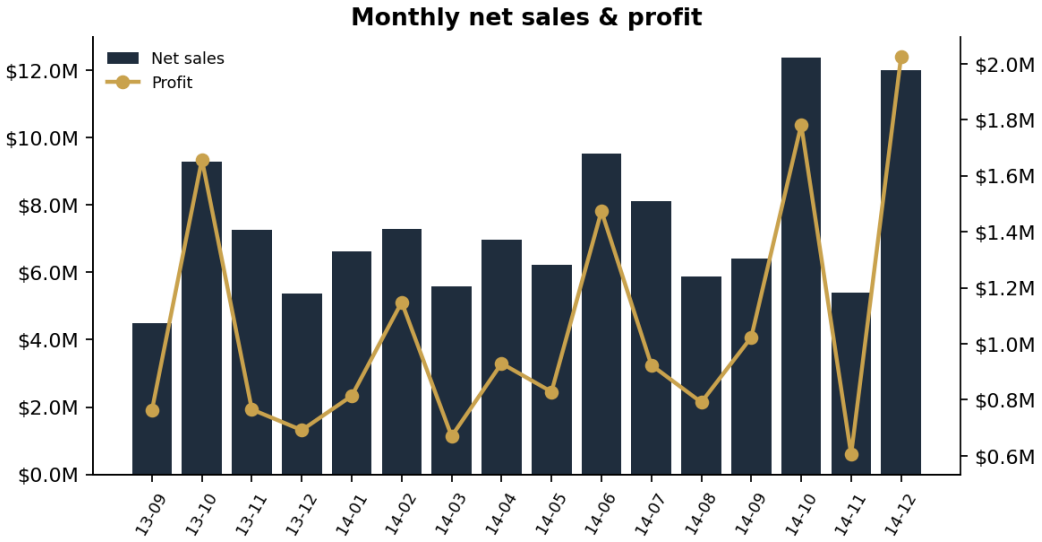
- Margin falls steadily as discounts deepen: 21.9% -> 17.9% -> 14.4% -> 9.1%.
- High-band deals are 35% of all deals and account for $5.3M of the $9.2M total discounts.
- All 58 loss-making deals sit in the Medium (25) and High (33) bands.

4.4 Country and product



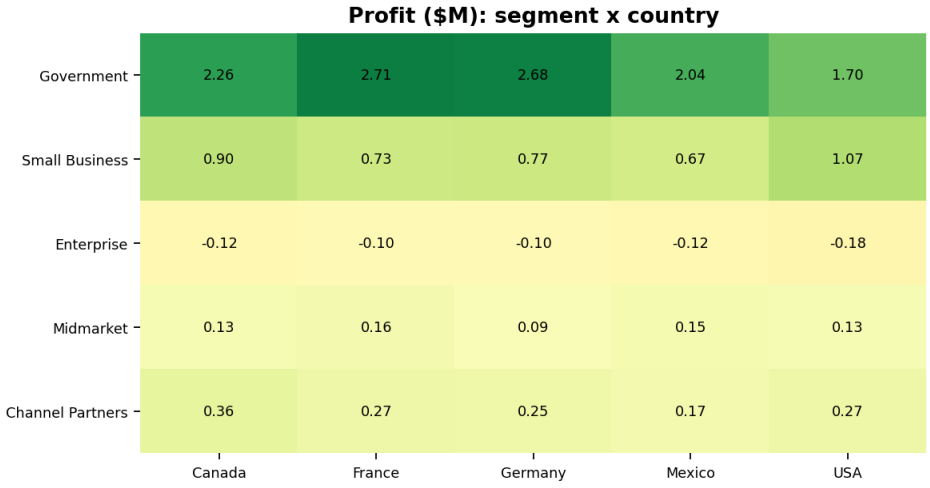
- USA: highest sales ($25.0M) but lowest margin (12.0%) and the highest discount rate (8.2%).
- France and Germany: best margins (15.5-15.7%) with the lowest discount rates (5.7-6.6%).
- Paseo is the top product (28% of sales, $4.8M profit). Product margins are close (12.6-15.9%), so segment and discounting matter more than product mix.

4.5 Trend and seasonality

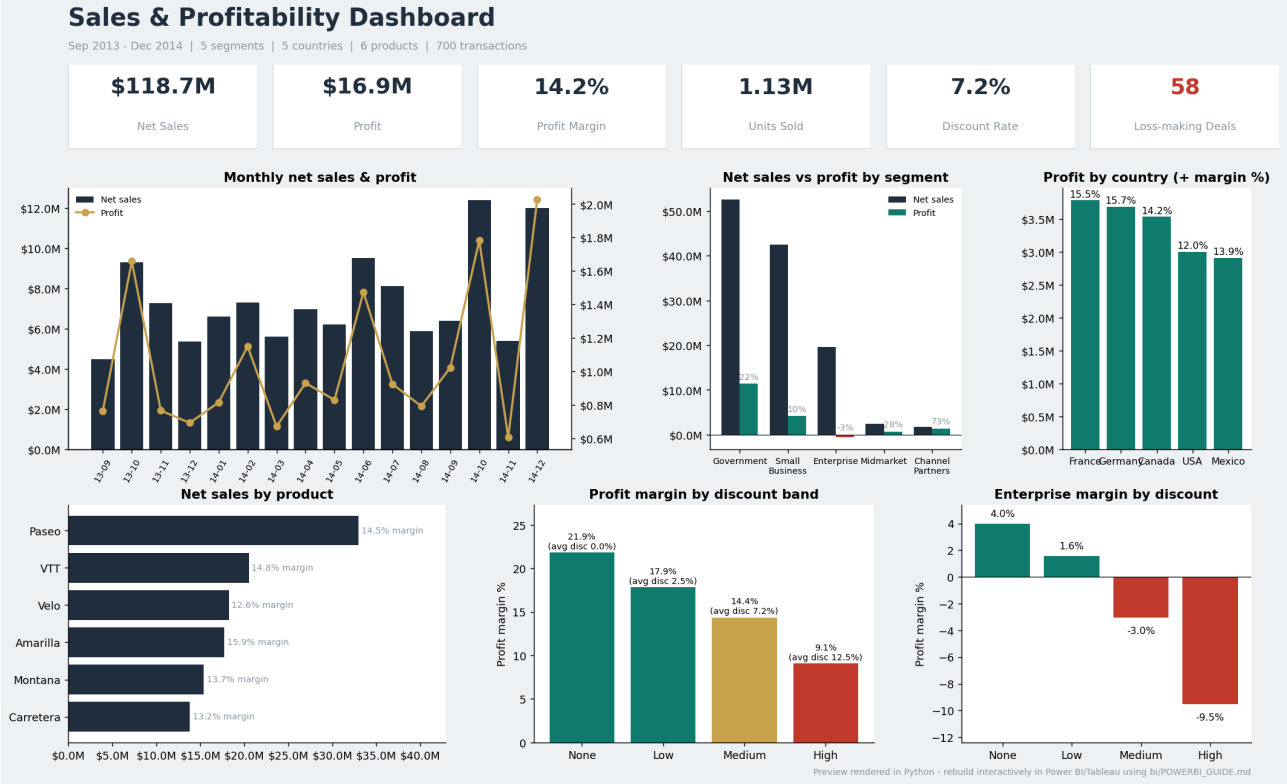


- October and December 2014 were the strongest months ($12.4M and $12.0M); November 2014 was weak ($5.4M).
- Like-for-like Sep-Dec: **$36.2M in 2014 vs $26.4M in 2013 (+37%)**; profit $5.4M vs $3.9M.

4.6 Segment x country profit



5. Dashboard



Dashboard layout: 6 KPI cards, trend, segment, country, product, discount and Enterprise views, with slicers for Year, Segment, Country, Product and Discount Band. The interactive version is *bi/dashboard.html*; Power BI steps and DAX measures are in *bi/POWERBI_GUIDE.md*.

6. Recommendations

- **Fix Enterprise pricing:** raise the 125 list price or cap Enterprise discounts at about 4% (the break-even point). Require approval for anything above.
- **Introduce a discount policy:** margin drops by half in the High band. Cap discounts for Small Business and Enterprise first; review whether High-band discounts actually lift volume enough.
- **Protect and grow Government** - it generates two-thirds of profit; concentration is also a risk, so diversify.
- **Grow Channel Partners and Midmarket** - 73% and 28% margins but under 4% of sales combined.
- **USA:** investigate why discounts are highest and margin lowest; align with France/Germany practices.
- **Plan for seasonality:** stock and staff for October and December peaks; look at why November dips.

7. Limitations

- Only 16 months of data and only 4 months in 2013 - trends and YoY growth should be treated with caution.
- The dataset has no customer, region-within-country or cost-breakdown fields, so causes (e.g. why Enterprise is priced at 125) cannot be confirmed from data alone.
- 'Manufacturing Price' is a per-product constant that does not reconcile with COGS, so COGS was used for all profit analysis.